UPI Power BI Project – Complete Guide

1. Project Objective

Build a complete Power BI dashboard for UPI transaction analysis and fraud detection.

The dashboard should help analyze:

- Transaction trends
- Fraud patterns
- Customer behavior
- Banking insights
- State-wise performance
- Time-based transaction activity
- Device and network usage

---

2. Dataset Understanding

Columns in Dataset

Column Name	Description
transaction id	Unique transaction ID
timestamp	Date and time of transaction
transaction type	P2P or P2M
merchant_category	Transaction category
amount (INR)	Transaction amount
transaction_status	SUCCESS / FAILED
sender_age_group	Sender age group
receiver_age_group	Receiver age group
sender_state	Sender state
sender_bank	Sender bank
receiver_bank	Receiver bank
device_type	Android / iOS
network_type	3G / 4G / 5G / WiFi
fraud_flag	0 = Normal, 1 = Fraud
hour_of_day	Transaction hour
day_of_week	Day name
is_weekend	Weekend flag

---

3. Data Cleaning / Data Treatment Questions

Null Value Checks

1. Check null values in all columns.
2. Replace blanks in merchant category.
3. Replace missing network type values.
4. Handle missing transaction status.

Duplicate Handling

5. Check duplicate transaction IDs.
6. Remove duplicate records.
7. Validate unique transaction IDs.

Data Type Corrections

8. Convert timestamp into DateTime format.
9. Convert amount column into numeric datatype.
10. Ensure fraud flag is integer.
11. Validate hour_of_day datatype.

Data Standardization

12. Standardize bank names.
13. Remove extra spaces.
14. Correct inconsistent state names.
15. Standardize merchant categories.
16. Correct device type naming.

Validation Checks

17. Verify amount is not negative.
18. Validate hour range (0–23).
19. Check fraud flag contains only 0 or 1.

20. Validate weekend flag.
21. Check transaction status values.

Outlier Detection

22. Identify high-value outlier transactions.
23. Detect suspicious repeated amounts.
24. Analyze unusual transaction timings.

---

4. New Columns to Create

Date Columns

Transaction Date

```
Transaction Date = DATEVALUE('UPI'[timestamp])
```

Year

```
Year = YEAR('UPI'[timestamp])
```

Month Name

```
Month = FORMAT('UPI'[timestamp], "MMMM")
```

Month Number

```
Month Number = MONTH('UPI'[timestamp])
```

Quarter

```
Quarter = "Q" & FORMAT('UPI'[timestamp], "Q")
```

Week Number

```
Week Number = WEEKNUM('UPI'[timestamp])
```

Day Name

```
Day Name = FORMAT('UPI'[timestamp], "dddd")
```

Time-Based Columns

Time Slot

```
Time Slot =
SWITCH(
    TRUE(),
    'UPI'[hour_of_day] < 12, "Morning",
    'UPI'[hour_of_day] < 17, "Afternoon",
    'UPI'[hour_of_day] < 21, "Evening",
    "Night"
)
```

Peak Hour Flag

```
Peak Hour =
IF(
    'UPI'[hour_of_day] >= 18 && 'UPI'[hour_of_day] <= 22,
    "Peak",
    "Non Peak"
)
```

Amount-Based Columns

Amount Bucket

```
Amount Bucket =
SWITCH(
    TRUE(),
    'UPI'[amount (INR)] < 500, "Low",
    'UPI'[amount (INR)] < 2000, "Medium",
    "High"
)
```

High Value Transaction

```
High Value Flag =
IF('UPI'[amount (INR)] > 5000, 1, 0)
```

Fraud Columns

Fraud Label

```
Fraud Label =
IF('UPI'[fraud_flag] = 1, "Fraud", "Normal")
```

Suspicious Time Flag

```
Suspicious Time =
IF(
    'UPI'[hour_of_day] >= 0 && 'UPI'[hour_of_day] <= 5,
    "Suspicious",
    "Normal"
)
```

Bank Analysis Columns

Same Bank Transaction

```
Same Bank =
IF(
    'UPI'[sender_bank] = 'UPI'[receiver_bank],
    "Yes",
    "No"
)
```

---

5. Measures (KPIs)

Total Transactions

```
Total Transactions = COUNT('UPI'[transaction id])
```

Total Amount

```
Total Amount = SUM('UPI'[amount (INR)])
```

Average Transaction Amount

```
Average Amount = AVERAGE('UPI'[amount (INR)])
```

Maximum Amount

```
Max Amount = MAX('UPI'[amount (INR)])
```

Minimum Amount

```
Min Amount = MIN('UPI'[amount (INR)])
```

Fraud Count

```
Fraud Count =
CALCULATE(
    COUNT('UPI'[transaction id]),
    'UPI'[fraud_flag] = 1
)
```

Fraud Percentage

```
Fraud % =
DIVIDE([Fraud Count], [Total Transactions]) * 100
```

Success Transactions

```
Success Transactions =
CALCULATE(
    COUNT('UPI'[transaction id]),
    'UPI'[transaction_status] = "SUCCESS"
)
```

Failed Transactions

```
Failed Transactions =
CALCULATE(
    COUNT('UPI'[transaction id]),
    'UPI'[transaction_status] = "FAILED"
)
```

Success Rate

```
Success Rate =
DIVIDE([Success Transactions], [Total Transactions]) * 100
```

Weekend Transactions

```
Weekend Transactions =
CALCULATE(
    COUNT('UPI'[transaction id]),
```

```
    'UPI'[is_weekend] = 1
)
```

Weekend Transaction %

```
Weekend Transaction % =
DIVIDE([Weekend Transactions], [Total Transactions]) * 100
```

---

6. Advanced DAX Measures

Running Total Amount

```
Running Total =
CALCULATE(
    [Total Amount],
    FILTER(
        ALLSELECTED('UPI'),
        'UPI'[timestamp] <= MAX('UPI'[timestamp])
    )
)
```

Fraud Growth %

```
Fraud Growth % =
DIVIDE(
    [Fraud Count] -
    CALCULATE(
        [Fraud Count],
        PREVIOUSMONTH('Calendar'[Date])
    ),
    CALCULATE(
        [Fraud Count],
        PREVIOUSMONTH('Calendar'[Date])
    )
)
```

State Rank by Amount

```
State Rank =
RANKX(
    ALL('UPI'[sender_state]),
    [Total Amount],
    ,
    DESC
)
```

---

7. Insight Questions for Dashboard

Transaction Analysis

1. Which state has the highest number of transactions?
2. Which bank handles maximum UPI transactions?
3. Which merchant category generates highest revenue?
4. Which day has maximum transactions?
5. Which month has highest transaction volume?
6. What is the peak transaction hour?
7. Which network type is used most?
8. Android vs iOS transaction comparison.
9. Which age group spends the most?
10. Which category has highest average transaction amount?
11. Which state has highest average amount?
12. Which time slot has maximum transaction activity?
13. Which bank combination occurs most?
14. Which category has maximum failed transactions?
15. Which device type is preferred?

Fraud Analysis

16. Which state has highest fraud rate?
17. Which bank has maximum fraud transactions?
18. Which merchant category is most risky?
19. Which network type has highest fraud rate?
20. Are frauds more common during night hours?
21. Weekend vs weekday fraud comparison.
22. Which device type is more vulnerable to fraud?
23. Which age group faces more fraud?
24. High-value fraud transaction analysis.
25. Monthly fraud trend analysis.
26. Which hour has maximum fraud transactions?
27. Which bank pair is most involved in fraud?
28. Which states show increasing fraud trend?
29. Fraud comparison between P2P and P2M.
30. What percentage of total amount is fraudulent?

Customer Behavior Analysis

31. Which age group prefers P2P transactions?
32. Which age group prefers shopping transactions?
33. Which states use 5G network most?
34. Which users use WiFi most frequently?
35. Which category is popular during weekends?
36. Which category is active at night?
37. Which banks are popular among young users?
38. Which users perform high-value transactions?
39. Spending pattern by age group.
40. Spending pattern by device type.

Banking Insights

41. Which sender bank has highest transaction amount?
42. Which receiver bank receives maximum payments?
43. Same-bank vs cross-bank transaction comparison.
44. Which bank has highest success rate?
45. Which bank has highest failed transaction rate?
46. Which bank processes high-value transactions most?
47. Which bank has most fraud during weekends?
48. Which bank is most used in P2M transactions?
49. Which bank has highest customer base?
50. Bank-wise fraud percentage comparison.

Time Analysis Questions

51. Morning vs evening transaction comparison.
52. Which day has highest fraud activity?
53. Which month has lowest transaction volume?
54. Which quarter generates maximum revenue?
55. Which hour has lowest success rate?
56. Weekend vs weekday amount comparison.
57. Monthly growth trend of transactions.
58. Peak shopping hours analysis.
59. Time slot analysis by transaction type.
60. Seasonal spending analysis.

---

8. Dashboard Pages

Page 1 — Executive Summary

Visuals:

- KPI Cards
- Total Amount Trend
- Fraud Percentage
- State-wise Transactions
- Bank-wise Revenue

Page 2 — Transaction Analysis

Visuals:

- Hourly Heatmap
- Merchant Category Analysis
- Day-wise Transactions
- Device Usage Analysis
- Network Type Distribution

Page 3 — Fraud Detection Dashboard

Visuals:

- Fraud Trend
- Fraud by State
- Fraud by Bank
- Fraud by Merchant Category
- Fraud by Time Slot

Page 4 — Customer Behavior Dashboard

Visuals:

- Age Group Analysis
- Spending Pattern
- Weekend Analysis
- P2P vs P2M Analysis
- Device Comparison

---

9. Recommended Visuals

Business Question	Recommended Visual
Transaction trend	Line chart
Fraud by state	Filled map
Bank comparison	Bar chart
Device usage	Donut chart
Fraud percentage	KPI Card
Time analysis	Heatmap
Category analysis	Treemap
Monthly comparison	Area chart
Bank hierarchy	Matrix
Customer behavior	Clustered bar chart

---

10. Advanced Power BI Features

1. Drill Through Pages
2. Dynamic Titles
3. Dynamic Tooltips
4. Bookmark Navigation
5. What-if Parameters
6. Conditional Formatting
7. Forecasting
8. Decomposition Tree
9. Key Influencer Visual
10. Row-Level Security
11. Smart Narrative
12. Field Parameters
13. Custom Theme
14. Buttons and Navigation
15. Interactive Filters

---

11. Suggested Filters / Slicers

1. Year
2. Month
3. State
4. Sender Bank
5. Receiver Bank
6. Merchant Category
7. Transaction Type
8. Device Type
9. Network Type
10. Fraud Flag
11. Age Group
12. Weekend Filter
13. Time Slot

---

12. Final Project Goals

After completing this project, you should be able to:

- Build end-to-end Power BI dashboards
- Create calculated columns using DAX
- Create advanced measures
- Perform fraud analytics
- Build interactive reports
- Use filters and slicers effectively
- Analyze banking transaction patterns
- Create professional dashboards for portfolio projects